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Task Description:	Space radiation poses a significant hazard to astronaut health both during the mission and at later times. However, uncertainty exists with respect to the level of risk to the CNS (central nervous system) following radiation doses relevant to those encountered during space missions. Radiation exposure in the brain affects the hippocampus, a structure critical for cognition. This proposal addresses section 1.2 of NASA Research Announcement NNJ12ZSA001N entitled, “Risk of Acute or Late Central Nervous System Effects from Space Radiation.” The proposed work focuses on changes in hippocampal network stability, including epigenetic mechanisms, in the development of cognitive dysfunction after space irradiation. Radiation exposure of the CNS significantly affects the hippocampus. Our preliminary data show impaired cognition in mice at two weeks following low dose space radiation. During spatial exploration, when an animal enters a specific location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory and resulting in the transcription of the immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc). Thus, analyzing Arc provides a reliable method to detect the number of active cells forming environment-specific maps and to capture the environmental specificity of place fields. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen across visits in different environments. The proposed work will focus initially on effects of space radiation on the stability of hippocampal networks, assessed by analyzing the expression of Arc and other genes important for memory formation. We hypothesize that the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability. Promoter region DNA methylation (5mC) of Arc leads to decreased expression and space radiation induces changes in DNA methylation. Moreover, DNA methylation plays a key role in regulating expression of genes required for spatial learning and memory. Thus, study of Arc expression and DNA methylation after charged particle exposure is warranted, followed by a more global search for DNA methylation changes in genes involved in learning and memory. A second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC), is abundant in the brain and found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. In this proposal, we will test the hypotheses that the detrimental effects of space radiation on cognition are associated with Arc-related changes in the hippocampal networks involved in learning and memory and that these changes are related to space radiation-induced perturbation of 5mC and/or 5hmC. Our approach will link assessment of hippocampus-dependent cognitive changes, hippocampal network stability, and alterations in global DNA methylation and expression of genes involved in learning and memory. In Specific Aim 1, we will test the hypothesis that low-dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on cognitive performance in object recognition and spatial learning and memory requiring navigation. In Specific Aim 2, we will test the hypothesis that low dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on hippocampal networks involving Arc, and that cognitive injury following space irradiation is associated with changes in hippocampal network stability. In Specific Aim 3, we will test the hypothesis that cognitive changes following space irradiation are associated with alterations in DNA methylation and expression of Arc and other genes involved in learning and memory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is relevant to NASA Integrated Research Plan and associated Research and Technology Gaps. Humans are exposed to ionizing irradiation under a variety of conditions associated with medical treatment. While the risks of developing cognitive impairments in those situations are fairly well established, uncertainty exists with respect to other types of radiation exposure, such as those experienced in the space environment. There is a critical need to establish the level of risk to the CNS following radiation doses relevant to those encountered during space missions. NASA has identified four categories of radiation risk in the Integrated Research Plan (IRP) and their associated research and technology gaps. This project is pertinent to 1.2 “Risk of Acute or Late Central Nervous System Effects from Space Radiation” and to Space Radiation Risks CNS1, CNS2, CNS6, Degenerative 1, and Degenerative 2. The project focuses on the role of changes in hippocampal network stability and epigenetic mechanisms involved in these changes in the development of cognitive dysfunction after space irradiation. CNS-related topics expressly targeted by the IRP and that are specifically or indirectly addressed in the current project include early and late effects of space irradiation on cognitive function and cellular and molecular biomarkers that might be associated with cognitive injury. Innovative technologies are being developed for integrated functional assays to assess effects of space radiation on brain function.
	This project focused on changes in hippocampal network stability and epigenetic mechanisms involved in the development of cognitive dysfunction after space irradiation. Radiation exposure of the central nervous system (CNS) can significantly affect the hippocampus, a structure critical for cognitive function. During spatial exploration, when an animal enters a specific location a distinct cell responds (known as the cell's place field). As a result of these place fields, hippocampal pyramidal neurons in the CA areas display robust high frequency activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory. Following visits to an environment for a second time, the same cells form stable neural networks and respond as a result of cellular activity induced by a learning experience. While place fields are highly reproducible and stable across multiple visits to the same environment, this stability is not seen in when different environments are visited. The detrimental effects of charged particles on hippocampus-dependent cognition are likely associated with changes in hippocampal networks. Analysis of the precise temporal kinetics of the plasticity-related immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc) using a method called catFISH (cellular compartment analysis of temporal activity using fluorescence in situ hybridization) provides an ideal and reliable approach for studying network stability. In a prior study we showed that Arc protein expression in the hippocampus is sensitive to brain-only exposure to 56Fe ions (600 MeV, 1 Gy). Arc expression is reduced with aging in the CA1 “place cells” within the mouse hippocampus and this reduction is correlated with increased promoter region DNA methylation (5mC). This is not surprising because changes in DNA methylation play a key role in regulating expression of genes required for spatial learning and memory. Gamma and space irradiation, including 56Fe ions and protons, induce changes in DNA methylation. Of significance for this project, recent studies show that a second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC) is abundant and stable in the brain. The TET (ten eleven translocation) enzymes convert 5mC to 5hmC, which likely play a role in gene regulation because 5hmC is often found at promoters of active genes. In contrast, 5mC is found at the promoters of inactive genes. Changes in 5mC, which are induced by environmental exposures including radiation and behavioral modification, play a role in processes including cancer and neurological disease. A similar role for 5hMC can be assumed due to its abundance in the brain, though 5hmC changes after radiation exposure and in neurological disease

has not yet been elucidated. We hypothesized that the detrimental effects of charged particles on hippocampus-dependent cognition are associated with Arc-related changes in the hippocampal networks involved in synaptic plasticity and memory. We further hypothesized that these changes are related to perturbation of 5mC and/or 5hMC in the hippocampus that affect Arc expression and expression of other learning and memory genes in the same signaling pathways and networks. To test these hypotheses, we used the following specific aims:
Aim 1: Determine short- and long-term effects of irradiation with 56Fe (600 MeV/n; 0.1, 0.2 or 0.4 Gy) or 28Si (600 MeV/n; 0.3, 0.6, or 0.9 Gy) ions or protons (150 MeV; 1 Gy) on novel object recognition and spatial learning and memory.

Aim 2: Determine the short- and long-term effects of irradiation with 56Fe (600 MeV/n; 0.1, 0.2 or 0.4 Gy) or 28Si (600 MeV/n; 0.3, 0.6, or 0.9 Gy) ions or protons (150 MeV; 1 Gy) on hippocampal networks mediating spatial information processing, and whether the detrimental effects of charged particle radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability.

Aim 3: Determine whether cognitive changes resulting from charged particle exposure are associated with altered DNA methylation (5mC and 5hMC) profiles for Arc and other genes involved in learning and memory in the same signaling pathways; and whether DNA methylation changes correlate with stable changes in gene expression.

New critical path questions that have resulted from the research:

What are the networks and pathways in the brain involved in effects of space radiation on cognition? Are the effects seen within 4 weeks after irradiation transient or not? Are the epigenetic effects seen following irradiation in the CNS specific for the brain or are similar effects seen in peripheral tissues? When we compare pathway changes following dose of 56Fe ion irradiation, 28Si ion irradiation, and proton irradiation that cause cognitive injury, is there a shared space radiation signature? If a shared space radiation signature exists, it would not only provide important insides in the mechanisms underlying cognitive injury following space irradiation and define which acute and long-term risks the CNS is exposed to most but it would also guide which compounds should be considered to mitigate the radiation effects based on existing FDA-approved compounds or even new compounds that are not FDA-approved yet.

In animal models, significant radiation-induced cognitive changes are seen without apparent histological changes in or around the hippocampus and surrounding white matter particularly after relatively low dose exposures. These results suggest a role for radiation-induced epigenetic changes in altered cognition because epigenetic changes can occur without altering cell structure; yet such changes can have large effects on gene expression. The cognitive effects of irradiation reflect impaired attention and memory more than low intellectual performance, supporting involvement of the hippocampus, a key component of the medial temporal lobe memory system. Recent published studies by us and others confirm the role of the hippocampus in radiation-induced cognitive injury after low or high LET (linear energy transfer) irradiation. Hippocampus-dependent spatial learning and memory is an important factor with respect to age-related cognitive decline and likely an important risk factor for performance during space missions. After high dose radiation exposures in rats, hippocampus-dependent learning and memory appeared less sensitive to stimuli. Whole body irradiation and sensitive hippocampus-dependent tests are predicted to detect effects of lower doses. In Specific Aim 1, we addressed the uncertainty required for risk assessment and determined short- and long-term effects of radiation with 56Fe or 28Si ions or protons on hippocampus-dependent cognitive performance. Work in later specific aims examined the role of radiation on network stability and epigenetic stability and how these changes relate back to the anticipated perturbations in cognitive performance.

Changes in hippocampal network stability and radiation-induced cognitive injury. Although space irradiation adversely impacts cognitive function, the cellular/molecular mechanisms responsible are not clear. We hypothesized that the detrimental effects of charged particles on hippocampus-dependent cognition are associated with changes in the hippocampal networks involved in synaptic plasticity and memory.

During spatial exploration, when an animal enters a specific spatial location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust high frequency activity. The place fields create an environment-specific "map." This map is thought to be critical in establishing the context of a memory. Following visits to the same environment, the same cells form stable neural networks as a result of cellular activity induced by a learning experience. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen in when different environments are visited. Analysis of the precise temporal kinetics of Arc using a method called catFISH (cellular compartment analysis of temporal activity using fluorescence in situ hybridization) provides a novel approach for the analysis of synaptic plasticity and neuronal function from a cellular and a network perspective. It allows studying network stability by detecting the number of active cells forming environment specific maps and capturing the environmental specificity of place fields. When control or sham-irradiated animals visit the same environment twice, ~90% of cells expressing Arc in the nucleus also express Arc in the cytoplasm, indicating that the same neurons were active during both explorations. In contrast, when two different environments are visited, the proportion of cells showing double labeling is very low. Arc has been used to provide important insight into the post-transcriptional infrastructure of gene expression involved in synaptic plasticity and memory. When neurons are engaged in information processing, Arc is rapidly transcribed and can be visualized and quantified after ~5 min. Subsequently, the mRNA is translocated to the cytoplasm where it remains detectable for ~20-30 min after the initial transcription. Ultimately, the mRNA is translocated to tagged synapses for protein synthesis. Synaptic tagging was first proposed to explain how neural signaling at a particular synapse creates a target for subsequent plasticity-related product (PRP) trafficking essential for sustained synaptic plasticity (either long-term potentiation or long-term depression). Thus, two different cellular compartments (nuclear and cytoplasmic) can be clearly distinguished, providing means to identify which neurons were active during distinct behavioral experiences. CatFISH provides accurate temporal and cellular resolution and facilitates mapping of neuronal activity across different hippocampal areas. This method provides a mechanistically relevant indicator of information processing related to memory and may reveal the cellular mechanisms underlying hippocampus-dependent cognitive injury.

Arc is an ideal IEG for studying hippocampal networks following irradiation. While other IEGs are involved in spatial information processing, Arc is the most responsive and the best characterized for kinetics of expression. Arc is an important mediator of information processing required for synaptic plasticity and memory and we showed that Arc expression in the hippocampus is sensitive to brain-only exposure to 56Fe ions (600 MeV, 1 Gy). In Specific Aim 2, we determined the short- and long-term effects of irradiation with 56Fe or 28Si ions or protons on hippocampal networks and whether the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability.

Task Progress:

Work in Specific Aim 3 provided additional information about altered expression of other genes involved in learning and memory in the same signaling pathways and networks as Arc with a more global screen for gene expression. Changes in DNA methylation affect the Arc promoter and play a role in learning and memory, hippocampal network stability, and radiation-induced cognitive injury. While it was originally assumed that DNA methylation patterns are stable and maintained for life, increasing evidence supports dynamic changes in DNA methylation over time in humans, nonhuman primates, and mice. Changes in DNA methylation are tissue-dependent, significantly conserved between humans and mice and play a key role in regulating expression of genes required for spatial learning and memory. In ensembles of CA1 place cells in the aged hippocampus, enhanced DNA methylation is proposed to contribute to the reduced Arc transcription. Because transcription of Arc is reduced following irradiation, we hypothesized that, as in the aging brain, there is enhanced methylation of the Arc promoter following radiation and that this contributes to reduced transcription. IEGs like Arc are poised for near-instantaneous transcription by the stalling of RNA polymerase II (Pol II) just downstream of the transcription start site. Promoter region methylation should affect the level of poised Arc transcripts. Effects of radiation on DNA methylation on Arc and other genes involved in learning and memory in the same signaling pathways and networks as Arc would have pathological consequences. Age-dependent changes in DNA methylation are thought to be associated with the risk to develop various diseases. Both hypomethylation and hypermethylation are implicated with neurological conditions.

Recently, our understanding of DNA methylation in mammals was expanded and confounded by the discovery of a hydroxy form, which is particularly abundant in the brain. Methylcytosine (5mC) is converted to hydroxymethylcytosine (5hmC) by the action of three TET enzymes (TET 1-3). Like 5mC, 5hmC levels and localizations are high and exceptionally dynamic in the brain during development and aging. Moreover, 5hmC is often found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. Several non-exclusive hypotheses have been offered to explain the presence of 5hmC in the mammalian genome, including control of gene expression and an intermediary in DNA demethylation. Although the exact role of 5hmC in the brain remains to be elucidated, the high and relatively stable levels suggest that demethylation is not a widespread phenomenon. Instead, its high level in brain neurons and inverse relation with 5hmC suggests strongly that 5hmC, like its precursor 5mC, plays an important a role in gene regulation. Therefore, any study of epigenetic change from radiation exposure must now include both forms of methylcytosine, even though determination of the exact role for 5hmC is ongoing. We assessed potential short-term and long-term effects of space radiation on both forms of methylcytosine in Aim 3 of this proposal.

As part of this project, we published a paper in March 2016 in BMC Genomics regarding the 20-week proton data demonstrating that proton irradiation induces persistent and tissue-specific DNA methylation changes in the left ventricle and hippocampus (see also Cumulative Bibliography hyperlink). We used an unbiased genome-wide DNA methylation study, involving ChIP-seq with antibodies to 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC) to identify DNA regions in which methylation levels have changed 22 weeks after a single exposure to proton irradiation. We used DIP-Seq to profile changes in genome-wide DNA methylation and hydroxymethylation following proton irradiation. In addition, we used published RNAseq data to assess whether differentially methylated regions were linked to changes in gene expression. The DNA methylation data showed tissue-dependent effects of proton irradiation and revealed significant major pathway changes in response to irradiation that are related to known pathophysiologic processes. Many regions affected in the ventricle mapped to genes involved in cardiovascular function pathways, whereas many regions affected in the hippocampus mapped to genes involved in neuronal functions. In the ventricle, increases in 5hmC were associated with decreases in 5mC. We also observed spatial overlap for regions where both epigenetic marks decreased in the ventricle. In hippocampus, increases in 5hmC were most significantly correlated (spatially) with regions that had increased 5mC, suggesting that deposition of hippocampal 5mC and 5hmC may be mechanistically coupled. These results demonstrate long-term changes in DNA methylation patterns following a single proton irradiation, that these changes are tissue specific, and that they map to pathways consistent with tissue specific responses to proton irradiation. Further, these results suggest novel relationships between changes in 5mC and 5hmC.

Although this was not part of the proposed experiments, we also use the same hippocampal samples for a metabolomics analysis. We finalized the initial analysis but as a new library became available to us, we are re-running these analyses and are planning a manuscript with these data. We are almost done verifying that all peaks ran on both a positive and negative charged column in the laboratory of our collaborator, Dr. Fred Stevens at the Linus Pauling Institute and College of Pharmacy at the Oregon State University. We plan to not only analyze metabolic pathway alterations in the left ventricle of the heart and hippocampus using metabolomics but also plan for an integrated analysis of DNA methylation and metabolomics, similar to what we recently reported in Scientific Report for hippocampal changes in apoE3 and apoE4 mice following a high-fat or low-fat diet (Johnson, L.A., Torres, E.R.S., Impey, S., Stevens, J.F., Raber, J. (2017) Apolipoprotein E4 and insulin resistance interact to impair cognition and alter the epigenome and metabolome. Scientific Reports, 7, 43701. Highlighted on the Alzheimer Research Forum and selected as paper of the month by the Oregon Health & Science University (OHSU) School of Medicine (SOM) for April 2017).

As part of this project, we also published in October 2016 a second paper in BMC Genomics (see Cumulative Bibliography hyperlink). This paper was regarding the 2- and 20-week 56Fe data. BMC Genomics selects articles that are of particular interest to the community to have a complimentary video summary created by their affiliates Research Square. Our article (Short- and long-term effects of 56Fe irradiation on cognition and hippocampal DNA methylation and gene expression) was selected to have a video created which is being promoted on the journal website and social media.

The youtube link is as follows: https:// ; The link that directs people to this publication is: (https://).

In this study, we examined object recognition in 6-month-old mice irradiated with 56Fe ions, a biological age more relevant to the typical ages of astronauts. Moreover, because the mechanisms mediating the detrimental effects of 56Fe ions on hippocampal function are unclear, we examined changes in hippocampal networks involved in synaptic plasticity and memory, gene expression, and epigenetic changes in cytosine methylation (5mC) and hydroxymethylation (5hmC) that could accompany changes in gene expression. We assessed the effects of whole body 56Fe ion irradiation at early (2 weeks) and late (20 weeks) time points on hippocampus-dependent memory and hippocampal network stability, and whether these effects are associated with epigenetic changes in hippocampal DNA methylation (both 5mC and 5hmC) and gene expression.

At the two-week time point, object recognition and network stability were impaired following irradiation at the 0.1 and 0.4 Gy dose, but not following irradiation at the 0.2 Gy dose. No impairments in object recognition or network stability were seen at the 20-week time point at any irradiation dose used. Consistent with this pattern, the significance of pathways for gene categories for 5hmC was lower, though not eliminated, at the 20-week time point compared to the 2-week time point. Similarly, significant changes were observed for 5mC gene pathways at the 2-week time point, but no significant gene categories were observed at the 20-week time point. Only the 5hmC changes tracked with gene expression changes. Thus, dose- and time-dependent epigenomic remodeling in the hippocampus following 56Fe ion exposure correlates with behavioral changes.

As part of this project, we also analyzed the 2- and 20-week proton data. We published a very data rich and relatively high impact paper with these data in Scientific Reports. In the paper, we did not only include the proton data but also included the bi-directional and shared epigenomic signatures following proton and 56Fe irradiation. Six-month-old mice were sham irradiated or irradiated with proton (150 MeV, 1Gy). For the novel object recognition test, the mice were habituated to an open field for three subsequent days. On day 4, the mice were exposed to the open field containing two objects. On day 5, the mice were exposed to the open field with two objects but one familiar object (present on day 4) was replaced by a novel one. The time spent with each object was recorded by video and hand-scored by a researcher blinded to the treatment of the mice.

We assessed whether the response of the hippocampus to proton irradiation is specific and prolonged and is linked to epigenetic mechanisms. We focused on the response of the hippocampus at early (2-weeks) and late (20-week) time points following whole body proton irradiation. We examined 5mC and 5hmC. Impairments in object recognition, spatial memory retention, and network stability following proton irradiation were observed at the two-week time point and correlated with altered gene expression and 5hmC profiles that mapped to specific gene ontology pathways. Significant overlap was observed between DNA methylation changes at the 2 and 20-week time points demonstrating specificity and retention of changes in response to radiation. Moreover, a novel class of DNA methylation change was observed following an environmental challenge (i.e., space irradiation), characterized by both increased and decreased 5hmC levels along the entire gene body. These changes were mapped to genes encoding neuronal functions including postsynaptic gene ontology categories. Thus, the brain’s response to proton irradiation is both specific and prolonged and involves novel remodeling of non-random regions of the epigenome.

Below, we describe the analysis of the 2-week 28Si data performed as part of our project. As described above, we recently reported that object recognition memory, which uses a 24-hour interval between learning and memory assessment to test hippocampal function, is impaired two weeks following irradiation of 6-month-old mice with 56Fe ions (600 MeV/n, 0.1 Gy) 22 and protons (150 MeV, 1 Gy). To determine the effects of 28Si ion irradiation on object recognition, 6-month-old mice were irradiated with 28Si ions (600 MeV/n, 0.3, 0.6, and 0.9 Gy) and cognitively tested two weeks later. The mice were first habituated to an open field for three subsequent days. 28Si ion irradiation did not affect activity levels in the open field over these three days. On day 4, the mice were exposed to the open field containing two objects for 15 min. On day 5, the mice were exposed to the open field with two objects for 15 min, but one familiar object (present on day 4) was replaced with a novel one. The percent time exploring the novel object was used to assess novel object recognition. All groups spent a comparable amount of time exploring the two identical objects. Following 28Si (600 MeV) ion irradiation, mice irradiated with 0.3 Gy were impaired, as revealed by no discrimination between the familiar and novel object, while those irradiated with higher doses were not. Consistent with this pattern of the dose-response curve, sham-irradiated mice and those irradiated with 56Fe (600 MeV) at 0.2 Gy showed novel object recognition but mice irradiated with either 0.1 Gy were impaired. This pattern is especially concerning as the lower doses would be expected to be more pertinent to doses astronauts will be exposed to during space missions.

The ability to recognize changes/novelty in the environment is critical for astronauts during space missions. For example, the inability to detect another ship close by was the reason for an unfortunate collision of two Navy boats. Object recognition testing in rodents is also highly translational to humans. We developed an object recognition test in humans based on the rodent test and showed that it is sensitive to detect cognitive impairments in children, adults, and the elderly. Recently, we incorporated eye tracking with object recognition testing and demonstrated that we can analyze novelty detection based on eye tracking responses, allowing assessment of cognitive performance of astronauts during space missions without them having to answer any specific questions about what they saw on the screen.

We next assessed whether the mechanisms mediating the cognitive effects of 28Si ion irradiation might be associated with reduced ability of hippocampal neurons to recognize similar environments and to discriminate distinct environments using the temporal kinetics of mRNA encoding activity-regulated cytoskeleton-associated protein (Arc). This ability was analyzed by assessing the percentage of Arc-positive neurons expressing both Arc-positive foci in the nucleus and Arc staining in the cytoplasm. This combination represents the population of neurons activated by both experiences. Exploratory behavior activated a comparable total number of neurons in sham-irradiated and irradiated mice in the CA1 and CA3 regions of the hippocampus. However, there was an effect of 28Si ion irradiation on the percent of neurons activated by both experiences. The percentage of Arc-positive neurons expressing Arc mRNA in the nucleus and cytoplasm in the CA3 region of the hippocampus of sham-irradiated mice was significantly higher following exposure twice to the same environment, as opposed to exposure to two different environments and there was a trend towards a difference in the CA1 region of the hippocampus of sham-irradiated mice, this was not seen in mice irradiated with 0.3 Gy that showed impaired cognitive performance. In mice irradiated with 0.9 Gy, who showed no impaired cognitive performance, there was a trend towards a difference in the CA3 region, but not the CA1 region. The CA1 and CA3 regions encode different information. The CA3 region is involved in rapid encoding of new detailed memories and bound to a large degree to a spatial coordinate system and more important for distinguishing different environments, while the downstream CA1 region can interpret this information from the CA3 region in the context of less spatially restricted information and is more important for recognizing the same environment. This suggests that in mice irradiated with 0.9 Gy the CA3 region might be less affected than the CA1 region and sufficient for detecting a novel object in the environment.

We next determined whether 28Si ion irradiation-induced cognitive injury is associated with pathway changes in the hippocampus by comparing hippocampal pathway changes in mice irradiated at a lower dose that caused cognitive injury with hippocampal pathway changes in mice irradiated at a higher dose that did not cause cognitive injury. Hippocampal changes in cytosine methylation, a major epigenetic modification involving the addition of a methyl group to cytosine (5mC), are involved in the regulation of expression of genes required for cognitive performance. A second form of DNA methylation, derived from 5mC by the action of three TET enzymes (TET 1-3), is hydroxymethylcytosine (5hmC). In brain, levels of TET2 are higher than those of TET1 or TET3. Therefore, TET2 is

believed most important for brain function. We did not see effects of 28Si ion irradiation on TET2 immunoreactivity in the dentate gyrus, CA1 or CA3 region of the hippocampus, or cortex. The levels of 5mC and 5hmC levels are high and exceptionally dynamic during brain development and aging, suggesting that they play critical roles. As described above, we reported alterations in hippocampal DNA methylation following proton and 56Fe ion irradiation. To determine the effects of 28Si ion irradiation on hippocampal DNA methylation, we generated cytosine methylation (5mC) and/or cytosine hydroxymethylation (5hmC)-DIP-Seq libraries from the hippocampi of sham irradiated mice and mice exposed to 0.3 or 0.6 Gy of 28Si ion radiation using highly specific antibodies. These libraries were sequenced to a depth of ~40 million tags of which 55-69% mapped to a genomic locus. Genomic regions enriched for 5mC or 5hmC were segmented using a Monte Carlo-based algorithm and the union of these regions was tested for significant difference using a negative binomial approach. This approach identified thousands of differentially methylated and hydroxymethylated regions (DMRs and DHRs, respectively), of which the majority were within 25 kb of a transcription start site.

These data show that 28Si ion radiation-induced cognitive injury is associated with hippocampal changes in DNA methylation of cytosine, a major epigenetic modification involving the addition or removal of methyl groups, and specific synaptic pathways. We next assessed whether there is a general space radiation synaptic signature and distinct changes in hippocampal DNA methylation comparing effects of 1) 56Fe and 28Si; 2) proton and 28Si; and 3) proton, 56Fe ion, and 28Si ion irradiation.

The general space radiation signatures included genes pertinent to Alzheimer's disease (AD), Amyotrophic lateral sclerosis (ALS), and Zellweger syndrome, also called cerebrohepatorenal syndrome. Zellweger syndrome is a congenital disorder characterized by the reduction or absence of functional peroxisomes in the cells of an individual. It is one of a family of disorders called leukodystrophies. Clearly, all three neurological conditions are severe, highlighting the CNS risk following space irradiation. Importantly, considering when the exposure was compared to when the hippocampal tissues were analyzed, the pathways are related to conditions we would anticipate to be occurring as late effects (following the mission) rather than acute effects, highlighting that space radiation exposure likely has long-term consequences as well.

The shared space radiation signature is also pertinent to the planned studies using sequential beam exposures as, based on our results, it would be expected that specific shared pathways might be particularly affected and these pathways could then be targeted for mitigation.

Bibliography Type:	Description: (Last Updated: 10/30/2023)
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